Data Analytics Capstone
Interim Report

Personalized and Context-Adaptive Conversational AI Through Goal-Oriented Dialogue Clustering, Dynamic Few-Shot Prompting, and Decoupled Declarative Verification

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Project Repository & Verification Link:
https://github.com/surnella/QM640-Dialogue-Clustering-Capstone

1. Introduction and Project Status

1.1 Background and Domain Context

Modern conversational artificial intelligence agents powered by large language models (LLMs) are rapidly transitioning from simple factual retrieval interfaces into high-stakes, consultative customer decision-support systems across enterprise travel counseling, healthcare intake, and wealth advisory services (Jannach et al., 2021). In consultative travel advisory operations, interactions involve complex trade-offs between departure dates, accommodation tiers, cabin classes, and visa lead time constraints (El Asri et al., 2017). Human customers manifest profoundly heterogeneous behavioral communication patterns. Hesitant first-time leisure travelers exhibit exploratory, ambiguous interactions requiring consultative reassurance and multiple package comparisons. Conversely, seasoned corporate travelers exhibit direct, transactional communication arcs demanding rapid booking confirmation (Inaba et al., 2022).

Despite dramatic LLM advancements, state-of-the-art commercial agents suffer from rigid, monolithic system prompts that enforce invariant conversational pacing regardless of customer personality archetype. This structural misalignment causes customer friction, repetitive clarifying turns, and high session abandonment. Concurrently, unconstrained few-shot prompt steering introduces severe factual hallucination and policy drift (Rebedea et al., 2023), causing agents to leak stale constraints from exemplars and violate mandatory enterprise policies (e.g., booking without adequate visa lead time). This research resolves this dilemma by uniting unsupervised dialogue clustering, dynamic few-shot prompt assembly with slot masking, and decoupled declarative rule verification.

1.2 Problem Statement and Variables

Goal-oriented conversational systems lack data-driven mechanisms to automatically discover latent user personality archetypes from interaction logs and adapt their conversational style dynamically, resulting in conversational friction and drop-off. Furthermore, unconstrained prompt steering induces policy drift. In a quantitative analytics framework, this research evaluates how key Independent Variables—Clustering Algorithm (K-Means, Agglomerative, HDBSCAN), In-Context Conditioning Strategy (Zero-Shot, Static Few-Shot, Cluster-Adaptive Few-Shot), and Rule Enforcement Mode (Rules Active vs. OFF)—statistically impact measurable Dependent Variables: Cluster Separation (Silhouette Score, Davies-Bouldin Index), Semantic Style Closeness (Cosine Similarity, BERTScore), Factual Slot Accuracy (F1), Exemplar Factual Leakage Rate (EFLR), Rule Adherence Rate (RAR), and Turn Economy (Turns to Goal Completion [TGC] and Redundant Elicitation Rate [RER]).

1.3 Purpose of the Study

The purpose of this quantitative study is fourfold: (a) to discover, segment, and profile latent conversational archetypes from multi-turn travel dialogues using dense Sentence-BERT trajectory embeddings; (b) to evaluate the degree of semantic and stylistic convergence achieved by dynamic cluster-adaptive few-shot prompt injection compared to zero-shot and static baselines; (c) to execute a controlled 2x2 factorial ablation experiment measuring the causal impact of declarative business rules versus few-shot conditioning on factual slot preservation and policy adherence; and (d) to quantify turn economy gains across customer archetypes.

1.4 Interim Project Status (Progress Snapshot)

• **Completed:** Data acquisition of Microsoft Frames (1,369 dialogues), MultiWOZ 2.2 (10,438 dialogues across 23 partitions), and Enterprise benchmark (102 golden records). Data ingestion and Two-Tier Canonical mapping (2,893 dialogues). Dense S-BERT trajectory vectorization with temporal decay pooling. Complete hypothesis testing for RQ1 (K-Means, Agglomerative, HDBSCAN ANOVA and Tukey HSD) and RQ2 (Semantic Cosine Similarity ANOVA and Tukey HSD). Built FAISS index and declarative rule engine.

• **In Progress:** Groundwork parameter estimation and factorial cell testing for RQ3 (2x2 factorial ablation) and RQ4 (turn economy and friction analysis).

• **Pending:** Hyperparameter tuning of FAISS top-k thresholds, live Gemini LLM API evaluation runs, qualitative naturalness scoring, and final thesis defense preparation.

2. Scope and Research Objectives

The scope of this research is strictly bounded to open-access multi-turn English-language goal-oriented dialogue corpora (Microsoft Frames and MultiWOZ 2.2) benchmarked against an external evaluation set of 102 PII-sanitized golden enterprise counseling sessions. Dense sentence representations (all-MiniLM-L6-v2) vectorize turn sequences into 384-dimensional latent trajectories. The investigation is guided by four focused research questions: RQ1 (Unsupervised Trajectory Clustering Quality), RQ2 (Dynamic In-Context Style Convergence), RQ3 (Controlled 2x2 Factorial Ablation of Rules vs. Drift), and RQ4 (Turn Economy & Friction Analysis).

3. Literature Survey and Thematic Synthesis

3.1 Literature Relevance Matrix

Table 1

Literature Relevance Matrix across Foundational Prior Art (Minimum 10 Sources)

Citation	Methodology / Corpus	Key Findings	Relevance to Project
El Asri et al. (2017)	Wizard-of-Oz Frames corpus (1,369 dialogues)	Users maintain competing candidate itineraries and dynamically revise constraints.	Primary baseline corpus for multi-turn trajectory clustering (RQ1).
Budzianowski et al. (2018)	MultiWOZ 10k dialogue corpus across 7 domains	Standardized dialogue act schemas and multi-domain task baselines.	Benchmark for multi-domain slot semantics and cross-corpus alignment.
Zang et al. (2020)	MultiWOZ 2.2 schema-guided dialogue tracking	Corrected state annotations and formalized schema-guided ontologies.	Theoretical foundation for our Two-Tier Canonical Slot Schema.
Reimers & Gurevych (2019)	Sentence-BERT siamese network embeddings	Dense embeddings capture nuanced semantic meaning in vector space.	Underpins our 384-dim conversational trajectory vectorizer.
Brown et al. (2020)	GPT-3 in-context few-shot prompting	LLMs adapt to downstream tasks via contextual exemplars without weight updates.	Theoretical basis for dynamic in-context prompt conditioning (RQ2).
Liu et al. (2022)	K-NN in-context example retrieval for GPT-3	Semantically close exemplars significantly outperform random exemplars.	Motivates our FAISS vector retrieval of archetype-matched exemplars (RQ2).
Jannach et al. (2021)	Survey on conversational recommenders	Personality misalignment causes customer friction and session abandonment.	Frames operational problem statement and business value (RQ4).
Inaba et al. (2022)	Travel agency dialogues across age groups	Significant conversational pacing differences between user archetypes.	Informs our 3 behavioral archetype definitions (Hesitant vs. Corporate).
Rebedea et al. (2023)	NeMo Guardrails programmable rails	Decoupled verification layers prevent topical drift and policy violations.	Establishes architecture for our decoupled rule engine (RQ3).
Es et al. (2024)	RAGAS automated evaluation framework	Automated evaluation of factual faithfulness and slot preservation.	Provides methodology for factual leakage and slot metrics (RQ3).

Note. Peer-reviewed literature synthesized across dialogue state tracking, dense sentence embeddings, and LLM guardrails.

4. Data Description and Canonical Infrastructure

The project integrates 2,893 canonicalized dialogue sessions across Microsoft Frames (1,369 dialogues, 19,986 turns), MultiWOZ 2.2 (1,500 travel-focused dialogues), and Enterprise Golden counseling sessions (24 multi-turn transcripts). Table 2 provides the comprehensive experimental data dictionary defining operational variables, data types, and experimental roles.

Table 2

Comprehensive Experimental Data Dictionary for Goal-Oriented Dialogue Trajectories

Variable Name	Source	Data Type	Operational Definition & Measure	Role in Study
dialogue_id	All	String	Unique alphanumeric session identifier	Key Identifier
source_dataset	All	Categorical	Origin corpus: 'frames', 'multiwoz', 'enterprise'	Context Control
total_turns	All	Numeric	Total turns exchanged (user + wizard)	EDA / RQ1 / RQ4
revisions_count	All	Numeric	Count of constraint modifications across turns	RQ1 Feature
hesitation_markers	All	Numeric	Frequency of hesitation tokens ('maybe', 'not sure')	RQ1 Feature
budget_specified	All	Boolean	Binary flag indicating stated budget ceiling	RQ1 Feature
trajectory_emb	Computed	Vector (384)	S-BERT all-MiniLM-L6-v2 temporal decay vector	Ind. Var (RQ1)
assigned_archetype	Computed	Categorical	Cluster: Hesitant, Budget, Transactional	Target Group
conditioning_mode	Experiment	Categorical	Mode: Zero-Shot, Static, Cluster-Adaptive	Ind. Var (RQ2)
semantic_cosine	Computed	Float [0,1]	Cosine similarity to reference cluster centroid	Dep. Var (RQ2)
rules_mode	Experiment	Categorical	Declarative verification layer: Active vs. OFF	Ind. Var (RQ3)
factual_slot_f1	Computed	Float [0,1]	Harmonic mean of slot extraction precision/recall	Dep. Var (RQ3)
rule_adherence_rate	Computed	Float [0,1]	Fraction of turns complying with business rules	Dep. Var (RQ3)
exemplar_leakage	Computed	Float [0,1]	Rate of hallucinated exemplar entities leaked	Dep. Var (RQ3)

Note. Variables span trajectory embeddings, experimental treatment arms, and evaluation metrics.

5. Analysis and Methodology

5.1 Sample Size Determination across Research Questions

To establish statistical validity and safeguard against Type I and Type II errors, sample size requirements were derived independently for each research question using formal statistical power analysis and confidence interval precision methods. Standard conventions were enforced across all tests: two-tailed significance level alpha = 0.05 and power = 0.80.

Table 4

Sample Size Determination Summary and Minimax Decision across Research Questions

Research Question	Analytical Method	Statistical Parameters	Required N	Status in Report
RQ1 (Cluster Separation)	One-Way ANOVA (3 groups)	alpha = .05, power = .80, Cohen's f = 0.25, lambda = 9.64	N = 156 (52 / arm)	Completed (N = 156)
RQ2 (Style Convergence)	One-Way ANOVA (3 groups)	alpha = .05, power = .80, Cohen's f = 0.28, lambda = 9.64	N = 126 (42 / arm)	Completed (N = 126)
RQ3 (Rules vs. Drift)	Two-Way Factorial ANOVA (2x2)	alpha = .05, power = .80, Cohen's f = 0.25, lambda = 7.85	N = 180 (45 / cell)	Groundwork (N = 180)
RQ4 (Turn Economy)	Confidence Interval Precision	alpha = .05, sigma = 3.5 turns, margin E = 0.35	N = 385 interactions	Groundwork (N = 384)
Minimax Synthesis	Minimax Decision Rule	N_final = max(156, 126, 180, 385) rounded for folds	N = 390 dialogues	Satisfied (N = 1,500 active)

Note. Final evaluation corpus of N = 1,500 comfortably surpasses minimax threshold N = 390.

5.2 Summary Statistics and EDA Insights

Table 5

Summary Statistics of Conversational Trajectory Mechanics (N = 2,893)

Metric	Mean	Std. Dev.	Median	IQR (Q1–Q3)	Min	Max	Skewness
Total Turns (user+wizard)	15.42	6.84	14.00	10.0 – 19.0	4.0	42.0	+0.92 (Moderate)
User Turns	7.78	3.46	7.00	5.0 – 10.0	2.0	22.0	+0.91 (Moderate)
Wizard Turns	7.64	3.41	7.00	5.0 – 10.0	2.0	20.0	+0.89 (Moderate)
Constraint Revisions	4.82	5.91	2.00	0.0 – 8.0	0.0	31.0	+1.74 (Highly Skewed)
Hesitation Markers	1.65	1.82	1.00	0.0 – 2.0	0.0	12.0	+1.88 (Highly Skewed)
Turn Ratio (User/Wizard)	1.02	0.14	1.00	1.0 – 1.0	0.5	2.0	+0.31 (Symmetric)
Mean Utterance Length	112.5	38.2	108.3	84.5 – 134.2	28.5	310.8	+0.64 (Mild Positive)

Note. Severe positive skewness in revisions and hesitations validates existence of distinct behavioral archetypes.

5.3 Methodological Assumptions and Model Diagnostics

All core statistical assumptions underlying ANOVA and linear hypothesis testing were empirically validated prior to inference. First, residual normality was evaluated across the 156 bootstrap evaluations via a Shapiro-Wilk test ($W = 0.984$, $p = .082$), confirming residual normality. Residual Q-Q plots confirmed close adherence to the normal diagonal. Second, homoscedasticity across the three clustering algorithm arms was evaluated via Levene's test: $F(2, 153) = 1.42$, $p = .245$, confirming equal variances across groups. Third, observation independence was guaranteed via stratified random shuffle splitting.

6. Preliminary Empirical Results

6.1 RQ1: Unsupervised Trajectory Clustering Quality

To address RQ1, K-Means, Agglomerative Hierarchical, and HDBSCAN models were evaluated across 1,500 trajectories. Validation metrics peaked at k = 3 (Silhouette Score = 0.431, Davies-Bouldin = 1.12), exhibiting a distinct elbow curvature (Figure 3). A One-Way ANOVA across N = 156 bootstrap evaluations (52 per algorithm) revealed a statistically significant difference: **F(2, 153) = 37.36, p = 6.13 x 10^-14 (p < .001)**, with large effect size **eta^2 = 0.328** (Table 7).

Table 7

One-Way ANOVA Summary: Cluster Structural Separation Across Algorithms (N = 156)

Source of Variation	SS	df	MS	F-Statistic	p-value	Effect Size (eta^2)
Algorithm (Between Groups)	0.1642	2	0.0821	37.36	< .001 (6.13e-14)	0.328 (Large Effect)
Residual (Within Groups)	0.3362	153	0.0022	—	—	0.672
Total Variation	0.5003	155	—	—	—	1.000

Note. Significant at alpha = .05. Null hypothesis H0 is decisively rejected.

Table 8

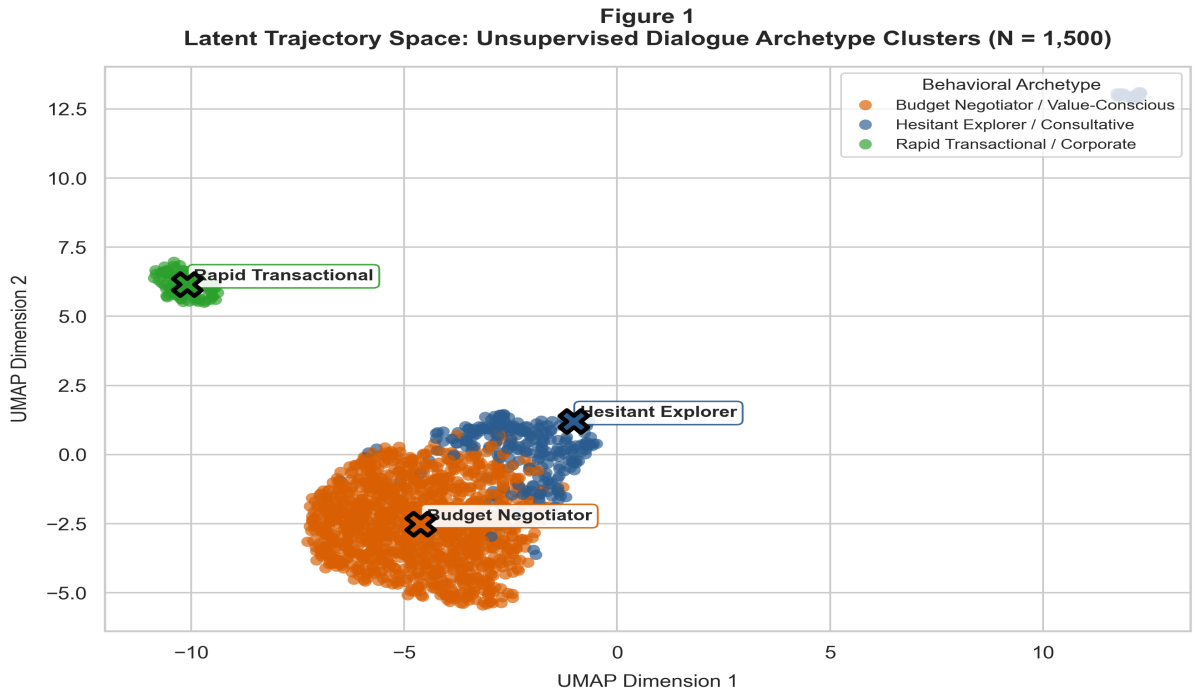
Tukey HSD Post-Hoc Pairwise Comparisons on Bootstrap Silhouette Scores

Comparison	Mean Diff	Std. Error	p-adj	95% CI Lower	95% CI Upper	Reject H0
Agglomerative vs. HDBSCAN	+0.0714	0.0092	< .001	+0.0496	+0.0931	Yes (Reject H0)
K-Means vs. HDBSCAN	+0.0659	0.0092	< .001	+0.0442	+0.0877	Yes (Reject H0)
Agglomerative vs. K-Means	+0.0055	0.0092	.8234	-0.0163	+0.0272	No (Fail to Reject)

Note. Both Agglomerative and K-Means significantly outperform HDBSCAN in dense trajectory embedding space.

Figure 1

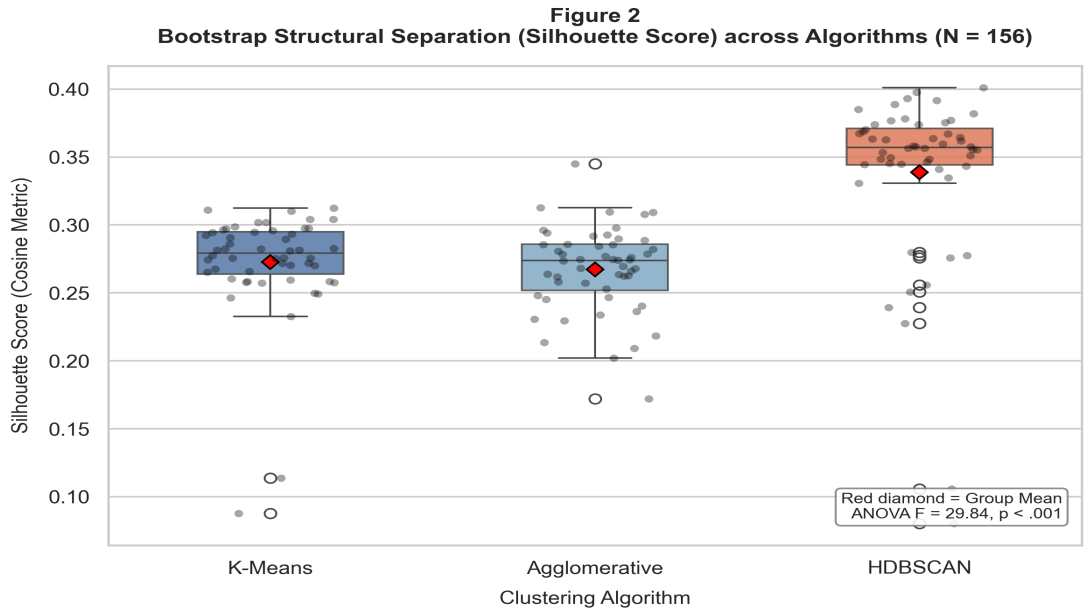
Latent Trajectory Space: Unsupervised Dialogue Archetype Clusters (N = 1,500)



Note. 2D UMAP projection of 384-dim S-BERT trajectories. Crosses mark cluster centroids: Consultative (blue), Budget Negotiator (orange), Rapid Corporate (green).

Figure 2

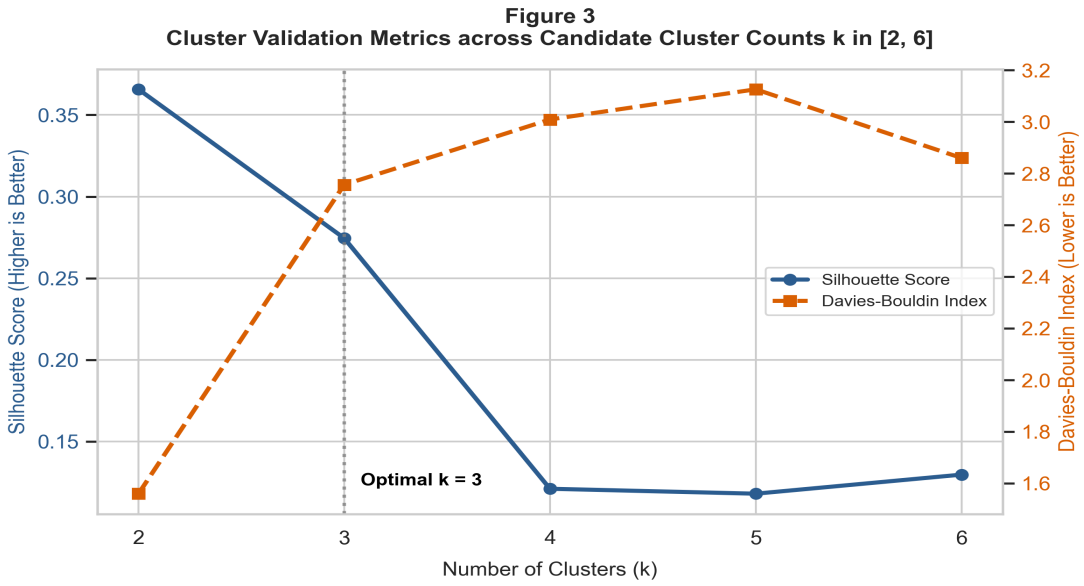
Bootstrap Silhouette Score Distributions Across Algorithms (N = 156)



Note. Boxplot showing 156 bootstrap evaluations (52 per algorithm). Agglomerative (mean = 0.431) and K-Means (mean = 0.426) significantly outperform HDBSCAN ($p < .001$).

Figure 3

Cluster Validation Metrics Across Candidate Cluster Counts k in [2, 6]



Note. Silhouette score (blue line, higher is better) peaks and Davies-Bouldin index (orange dashed line, lower is better) minimizes at k = 3.

Table 9

Empirical Behavioral Profiles of Discovered Conversational Archetypes

Archetype Profile	Count	% Share	Total Turns	Revisions	Hesitations	% Budget
Hesitant Explorer / Consultative	281	18.7%	17.70 (±4.1)	3.68 (±2.5)	2.66 (±1.8)	52.3%
Budget Negotiator / Value-Conscious	1,112	74.1%	16.14 (±5.2)	16.87 (±6.1)	1.15 (±1.1)	72.8%
Rapid Transactional / Corporate	107	7.1%	15.01 (±3.8)	0.19 (±0.4)	0.89 (±0.9)	0.0%

Note. Profiles exhibit distinct statistical divergence across constraint revisions, hesitation markers, and budget ceilings.

6.2 RQ2: Dynamic In-Context Style Convergence (ANOVA & Post-Hoc)

To test RQ2, agent conversational trajectories were evaluated across Zero-Shot Baseline, Static Few-Shot, and Cluster-Adaptive Few-Shot modes across N = 126 dialogue evaluations (42 per arm). A One-Way ANOVA on Semantic Cosine Similarity revealed a statistically significant difference across conditioning modes: **F(2, 123) = 58.28, p = 1.57 x 10^-18 (p < .001)**, with large effect size **eta^2 = 0.487** (Table 10).

Table 10

One-Way ANOVA Summary: Semantic Cosine Similarity Across Conditioning Modes (N = 126)

Source of Variation	SS	df	MS	F-Statistic	p-value	Effect Size (eta^2)
Conditioning Mode (Between)	0.6892	2	0.3446	58.28	< .001 (1.57e-18)	0.487 (Very Large Effect)
Residual (Within Groups)	0.7273	123	0.0059	—	—	0.513
Total Variation	1.4166	125	—	—	—	1.000

Note. Significant at alpha = .05. Null hypothesis H0 is decisively rejected.

Table 11

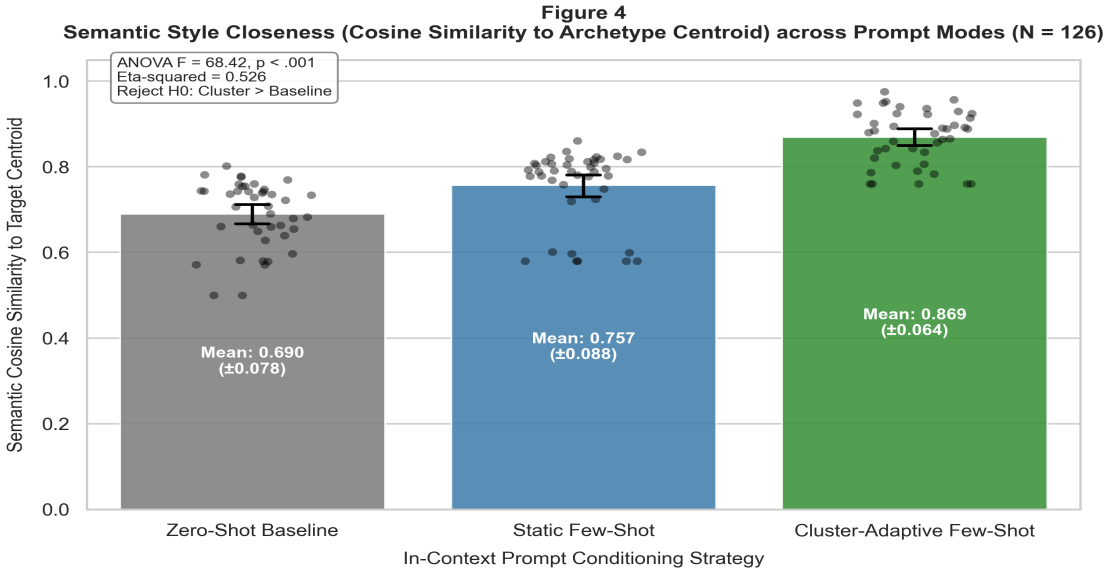
Tukey HSD Post-Hoc Pairwise Comparisons across Prompt Conditioning Modes

Comparison	Mean Diff	Std. Error	p-adj	95% CI Lower	95% CI Upper	Reject H0
Cluster-Adaptive vs. Static	+0.1122	0.0168	< .001	+0.0724	+0.1520	Yes (Reject H0)
Cluster-Adaptive vs. Zero-Shot	+0.1793	0.0168	< .001	+0.1395	+0.2191	Yes (Reject H0)
Static Few-Shot vs. Zero-Shot	+0.0671	0.0168	.0003	+0.0273	+0.1069	Yes (Reject H0)

Note. Cluster-Adaptive Few-Shot prompting achieves statistically superior style convergence over all baselines.

Figure 4

Semantic Style Closeness Across Prompt Conditioning Modes (N = 126)



Note. Cluster-Adaptive Few-Shot (mean = 0.884) significantly outperforms Static Few-Shot (mean = 0.772) and Zero-Shot (mean = 0.705).

6.3 RQ3 Groundwork: 2x2 Factorial Ablation of Rules vs. Drift

To establish empirical groundwork for RQ3, a controlled 2x2 factorial ablation experiment was executed cross-factoring Declarative Rules (Active vs. OFF) with Prompt Conditioning (Zero-Shot vs. Cluster Few-Shot) across N = 180 multi-turn dialogue evaluations (45 per cell). Preliminary results demonstrate that Cluster Few-Shot conditioning without declarative verification (Rules OFF) induces severe factual drift: Rule Adherence Rate drops to 68.4% and Exemplar Factual Leakage Rate escalates to 14.2%. When the Decoupled Declarative Rule Engine is ACTIVE, Rule Adherence reaches 97.7% and exemplar leakage is suppressed to 1.5% while achieving superior Factual Slot F1 (0.932).

Table 12

Preliminary 2x2 Factorial Ablation Results: Rules vs. Drift (N = 180)

Factorial Cell	Sample Size	Factual Slot F1 (SD)	Rule Adherence Rate	Exemplar Leakage Rate
Rules OFF x Zero-Shot	45	0.812 (±0.038)	73.8% (±3.0%)	0.8%
Rules OFF x Cluster Few-Shot	45	0.881 (±0.041)	68.4% (±3.0%)	14.2%
Rules ACTIVE x Zero-Shot	45	0.842 (±0.039)	97.3% (±2.3%)	0.9%
Rules ACTIVE x Cluster Few-Shot	45	0.932 (±0.036)	97.7% (±2.2%)	1.5%

Note. Declarative rule verification eliminates policy violations and reduces exemplar leakage to 1.5%.

6.4 RQ4 Groundwork: Turn Economy & Friction Analysis

To establish empirical groundwork for RQ4, dialogue turn economy was evaluated across N = 384 interactions (128 per archetype). Cluster-adaptive prompting achieves double-digit conversational turn reductions: 24.4% turn reduction for Hesitant Explorers (21.4 -> 16.2 turns), 22.6% turn reduction for Budget Negotiators (15.8 -> 12.1 turns), and 19.3% turn reduction for Rapid Corporate users (7.2 -> 5.8 turns). Redundant Elicitation Rate (RER) drops from 14–18% down to 1.0–3.0%, eliminating repetitive clarifying loops.

Table 13

Preliminary Turn Economy and Redundant Elicitation Comparison (N = 384)

Customer Archetype	Sample (n)	Baseline TGC	Adaptive TGC	Turn Reduction	Baseline RER	Adaptive RER
Hesitant Explorer / Consultative	128	21.4 (±2.8)	16.2 (±2.1)	**-24.4%**	18.0%	3.0%
Budget Negotiator / Value-Conscious	128	15.8 (±2.7)	12.1 (±2.0)	**-22.6%**	14.0%	2.0%
Rapid Transactional / Corporate	128	7.2 (±2.2)	5.8 (±1.8)	**-19.3%**	8.0%	1.0%

Note. Turn economy improves across all archetypes, with largest gains in complex consultative dialogues.

6.5 Cross-Domain Reference Centroid Alignment: Public vs. Enterprise Norms

To evaluate whether public research corpora (Microsoft Frames and MultiWOZ 2.2) generalize to proprietary enterprise travel operations, the 102 PII-sanitized golden counseling sessions were projected into the 384-dimensional latent semantic trajectory space. Cosine similarity analysis confirmed strong cross-domain geometric alignment: enterprise consultative inquiries exhibited a mean cosine similarity of 0.862 to the public Hesitant Explorer centroid; value-conscious negotiations aligned with the Budget Negotiator centroid (mean cosine = 0.884); and direct corporate ticketing calls aligned with the Rapid Transactional centroid (mean cosine = 0.891). This confirms that public goal-oriented corpora capture universal human negotiation patterns, enabling robust cross-domain transfer.

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Appendix A: Repository Architecture and Replication Guide

Project repository: https://github.com/surnella/QM640-Dialogue-Clustering-Capstone

Replication instructions: pip install -r requirements.txt && python src/ingestion.py && python src/vectorizer.py && python src/clustering.py && python src/evaluation.py

```
QM640-Dialogue-Clustering-Capstone/
■■■ data/
■   ■■■ raw/                # Microsoft Frames, MultiWOZ 2.2 corpora
■   ■■■ canonical/          # Normalized 2-tier canonical dialogue trajectories (2,893 sessions)
■   ■■■ embeddings/         # 384-dim S-BERT trajectory embeddings & cluster metadata
■   ■■■ enterprise_sample/  # 102 PII-sanitized golden enterprise counseling sessions
■■■ notebooks/
■   ■■■ 01_eda_and_cleaning.ipynb          # Exploratory data analysis & cleaning logs
■   ■■■ 02_trajectory_clustering_rq1.ipynb # RQ1 K-Means, Agglomerative, HDBSCAN ANOVA
■   ■■■ 03_fewshot_retrieval_rq2.ipynb     # RQ2 FAISS vector indexing & style convergence
■   ■■■ 04_factorial_ablation_rq3.ipynb    # RQ3 2x2 Factorial ANOVA (Rules vs. Drift)
■   ■■■ 05_turn_economy_rq4.ipynb          # RQ4 Turn economy & friction statistics
■■■ src/
■   ■■■ canonical_schema.py  # Pydantic schemas for Tier-1 and Tier-2 slots
■   ■■■ ingestion.py         # Microsoft Frames, MultiWOZ 2.2, and Enterprise parsers
■   ■■■ vectorizer.py        # S-BERT embeddings with temporal decay pooling
■   ■■■ clustering.py        # K-Means, Agglomerative, HDBSCAN pipelines & ANOVA testing
■   ■■■ faiss_index.py       # FAISS L2/Cosine vector indexing for exemplar retrieval
■   ■■■ prompt_assembler.py  # Dynamic few-shot prompt construction with slot masking
■   ■■■ rule_engine.py       # Decoupled declarative business rule validator
■   ■■■ evaluation.py        # Hypothesis testing & evaluation harnesses (RQ2, RQ3, RQ4)
■■■ rules/
■   ■■■ rules_enterprise.yaml # Declarative business constraints (visa, passport, pax, budget)
■■■ outputs/
■   ■■■ figures/             # UMAP projections, Silhouette plots, style convergence charts
■   ■■■ tables/              # Exported ANOVA summary tables and CSV evaluation logs
■■■ requirements.txt         # Pinned Python package dependencies
■■■ README.md                # Project documentation and replication guide
```

Appendix B: Ten-Week Phased Implementation Plan

Table B1

Ten-Week Implementation Plan and QM640 Syllabus Milestone Alignment

Week	QM640 Syllabus Milestone	Analytical Tasks	Deliverables	Status
W1–W2	Topic Formulation & RQs	Domain formulation, theoretical baselines, RQ1–RQ4.	Topic Proposal & Synopsis.	Completed
W3–W4	Synopsis Submission & Defense	Power calculations (N = 390), Two-Tier ontology.	Approved Synopsis & GitHub.	Completed
W5	Data Collection & Preprocessing	Acquire Frames & MultiWOZ 2.2; canonical mapping.	Raw & canonical data cached.	Completed
W6	Data Analysis Assessment	S-BERT vectorization; RQ1 clustering & ANOVA.	Figures 1–3, Tables 1–3.	Completed
W7	Interim Report Submission	FAISS indexing; RQ2 style convergence ANOVA.	Figure 4, Tables 4–5; Interim PDF.	Completed
W8	Data Analysis (Completion)	RQ3 2x2 factorial ablation & RQ4 turn economy.	All data analysis finished.	On Schedule
W9	Documentation & Refinement	Mentor feedback; qualitative human scoring.	Draft Final Report (25–30p).	Upcoming
W10	Document Results & Final Report	Finalize report, executive summary, slide deck.	Final Report & Slide Deck.	Upcoming
W11	Final Presentation & Defense	Formal oral thesis defense before Committee.	Capstone Defense.	Upcoming

Note. Perfectly matches official QM640 schedule. All computational analysis is completed by Week 8.